Extra Practice 1: Mixed and Entire Radicals

A. Skills

1. Without a calculator, evaluate where possible.

(a) $\sqrt{196}$ (b) $-\sqrt[3]{1000}$ (c) $\sqrt{-4}$ (d) $\sqrt[4]{16}$ (e) $\sqrt[3]{-\frac{1}{27}}$ (f) $\sqrt{0.16}$

2. Classify each as rational or irrational. (a) $\sqrt{121}$ (b) $\sqrt{120}$ (c) $\sqrt[3]{-125}$ (d) $\sqrt[3]{100}$

3. Write each radical as a mixed radical in simplest form.

(a) $\sqrt{32}$ (b) $\sqrt{108}$ (c) $\sqrt{175}$ (d) $2\sqrt{72}$

(e) $\sqrt[3]{40}$ (f) $\sqrt[3]{-24}$ (g) $\sqrt[3]{128}$ (h) $\sqrt[4]{80}$

4. Write each mixed radical as an entire radical.

(a) $5\sqrt{3}$ (b) $9\sqrt{2}$ (c) $-6\sqrt{3}$ (d) $4\sqrt[3]{2}$ (e) $-2\sqrt[3]{9}$ (f) $3\sqrt[4]{2}$

B. Applying

5. Without a calculator, arrange from least to greatest: $2\sqrt{15},\ 4\sqrt{3},\ 3\sqrt{5},\ \sqrt{58},\ 6\sqrt{2}$

6. Without a calculator, arrange from greatest to least: $3\sqrt[3]{2},\ 2\sqrt[3]{7},\ \sqrt[3]{50},\ -3\sqrt[3]{2},\ -\sqrt[3]{60}$

7. A square courtyard has an area of 363 m^2. Determine the exact side length and perimeter in simplest form.

8. Determine the exact distance between $(-3, 4)$ and $(3, 1)$ in simplest form.

9. Given $\sqrt{2} \approx 1.414$, approximate without a calculator: (a) $\sqrt{200}$ (b) $\sqrt{0.02}$ (c) $\sqrt{18}$ (d) $\sqrt{800}$

10. Which list is in order from least to greatest?

A. $2\sqrt{5}, 3\sqrt{2}, 2\sqrt{3}$ **B.** $2\sqrt{3}, 3\sqrt{2}, 2\sqrt{5}$ **C.** $3\sqrt{2}, 2\sqrt{5}, 2\sqrt{3}$ **D.** $2\sqrt{5}, 2\sqrt{3}, 3\sqrt{2}$

11. Numerical Response When $3\sqrt[3]{7}$ is written as the entire radical $\sqrt[3]{n}$, the value of n is _____.

C. Stretch

12. Write $\sqrt{288}$ in the form $a\sqrt{b}$ in every possible way with whole numbers $a > 1$ and $b > 1$. Which one is the simplest form?

13. Determine the largest whole number k for which $k\sqrt{5} < 30$.

14. Without a calculator, decide which is greater, $\sqrt[3]{5}$ or $\sqrt{3}$. (Hint: raise each number to the 6th power.)

Extra Practice 2: Radicals with Variables and Their Restrictions

A. Skills

1. State the restriction on the variable.
(a) $\sqrt{x-2}$ (b) $\sqrt{4x+12}$ (c) $\sqrt{15-5x}$

(d) $\sqrt[3]{6-x}$ (e) $\sqrt[4]{2x}$ (f) $\sqrt{-x-4}$

2. Simplify. Assume variables under an even index are ≥ 0.
(a) $\sqrt{x^{14}}$ (b) $\sqrt{a^9}$ (c) $\sqrt[3]{b^{10}}$ (d) $\sqrt{32x^3}$

(e) $\sqrt{125a^6b^5}$ (f) $\sqrt[3]{-24m^9}$ (g) $\sqrt[3]{500x^4y^2}$ (h) $\sqrt[4]{32a^6}$

3. Write each mixed radical as an entire radical. Assume variables under an even index are ≥ 0.
(a) $6x\sqrt{x}$ (b) $3a\sqrt{2a}$ (c) $-4m^2\sqrt[3]{m}$ (d) $2xy\sqrt{5y}$

B. Applying

4. Consider $\sqrt{3x+6}$. State the restriction, then evaluate exactly in simplest form where possible: $x=1$, $x=-1$, $x=14$, $x=-4$.

5. A square has an area of $72x^8$ square units. Determine its side length and perimeter in simplest form.

6. A cube has a volume of $250a^5$ cubic units. Determine its edge length in simplest form.

7. Pat wrote $\sqrt{18x^5} = 3x^4\sqrt{2x}$, $x \geq 0$. Explain the error and correct it.

8. Which radical is defined when $x=-3$?
A. $\sqrt{x+2}$ **B.** $\sqrt{-2x}$ **C.** $\sqrt[4]{x}$ **D.** $\sqrt{x-1}$

9. Numerical Response When $2a^3\sqrt[3]{3a}$ is written as the entire radical $\sqrt[3]{ka^m}$, the value of $k+m$ is _____.

C. Stretch

10. For what values of x is $\sqrt{x-2}+\sqrt{6-x}$ defined?

11. Simplify $\sqrt{8(x-1)^3}$, $x \geq 1$. (Treat $x-1$ as a single base.)

12. For which values of x is $\sqrt{(x-3)^2} = x-3$ true? Test $x=1$ to explain.

13. A square has an area of $27x^5$ square units, $x \geq 0$.

(a) Write its side length in simplest form.

(b) Use your answer to find the side length when $x=3$. Check with the area.

Extra Practice 3: Adding and Subtracting Radicals

A. Skills

1. Simplify. State the restriction in (e).

(a) $9\sqrt{10} - 2\sqrt{10}$ (b) $-\sqrt{6} + 5\sqrt{6} - 3\sqrt{6}$ (c) $3\sqrt[3]{7} + 8\sqrt[3]{7}$

(d) $2\sqrt{5} + 6\sqrt{3} - 7\sqrt{5} + \sqrt{3}$ (e) $12\sqrt{x} - 5\sqrt{x}$ (f) $3\sqrt[3]{4} + 2\sqrt{4}$

2. Write each expression as a single radical in simplest form.

(a) $\sqrt{12} + \sqrt{75}$ (b) $\sqrt{99} - \sqrt{44}$ (c) $2\sqrt{45} + \sqrt{80}$

(d) $\sqrt{162} - 2\sqrt{32} + \sqrt{8}$ (e) $\sqrt[3]{24} + \sqrt[3]{81}$ (f) $\sqrt[3]{-250} + 4\sqrt[3]{16}$

3. Simplify.

(a) $\sqrt{28} + \sqrt{12} - \sqrt{63} + \sqrt{48}$ (b) $\sqrt{500} - \sqrt{180} + \sqrt{242}$

(c) $\sqrt[4]{96} + \sqrt[4]{486}$ (d) $\frac{1}{3}\sqrt{72} + \frac{2}{3}\sqrt{50}$

B. Applying

4. Simplify. State the restrictions on the variables.

(a) $\sqrt{36b} - \sqrt{9b}$ (b) $\sqrt{20x^3} + x\sqrt{45x}$

(c) $\sqrt[3]{-54a^4} + a\sqrt[3]{2a}$ (d) $\sqrt{25x + 50} - \sqrt{x + 2}$

5. An isosceles triangle has two equal sides of length $\sqrt{50}$ cm and a base of length $\sqrt{72}$ cm. Determine its perimeter in simplest form.

6. A triangle has vertices $(0, 0)$, $(3, 3)$ and $(5, -1)$. Determine its perimeter in simplest form.

7. A rectangle has length $\sqrt{75}$ m and perimeter $14\sqrt{3} + 4\sqrt{2}$ m. Determine its width in simplest form.

8. Which expression is *not* equal to $6\sqrt{2}$?

A. $\sqrt{72}$ **B.** $\sqrt{18} + \sqrt{18}$ **C.** $\sqrt{50} + \sqrt{2}$ **D.** $\sqrt{36} + \sqrt{2}$

9. Numerical Response When $3\sqrt{28} + \sqrt{175} - \sqrt{63}$ is written in the form $a\sqrt{7}$, the value of a is _____.

C. Stretch

10. (a) Show that $\sqrt{3} + \sqrt{12} = \sqrt{27}$, even though $3 + 12 \neq 27$.

(b) Determine the whole number k for which $\sqrt{5} + \sqrt{k} = \sqrt{80}$.

11. List every whole number k from 1 to 100 for which $\sqrt{12} + \sqrt{k}$ can be written as a single radical.

12. In an arithmetic sequence, $t_1 = \sqrt{12}$ and $t_5 = \sqrt{300}$. Determine d and t_3 in simplest form.

Extra Practice 4: Multiplying Radicals

A. Skills

1. Multiply and simplify.

(a) $\sqrt{3} \cdot \sqrt{13}$

(b) $(4\sqrt{5})(-3\sqrt{2})$

(c) $\sqrt{6} \cdot \sqrt{24}$

(d) $(2\sqrt{14})(5\sqrt{2})$

(e) $(-\sqrt{15})(-4\sqrt{10})$

(f) $\left(3\sqrt[3]{6}\right)\left(\sqrt[3]{36}\right)$

(g) $\left(6\sqrt{2}\right)^2$

(h) $-\left(3\sqrt[3]{2}\right)^3$

(i) $\left(\sqrt[3]{25}\right)\left(2\sqrt[3]{10}\right)$

2. Expand and simplify.

(a) $\sqrt{7}\left(3-\sqrt{7}\right)$

(b) $4\sqrt{2}\left(\sqrt{6}+2\sqrt{10}\right)$

(c) $\sqrt{5}\left(\sqrt{15}-2\sqrt{5}+\sqrt{10}\right)$

3. Simplify. State the restrictions on the variables.

(a) $\left(3\sqrt{2x}\right)\left(\sqrt{8x}\right)$

(b) $\sqrt{5a}\left(\sqrt{5a}-2\right)$

(c) $\left(\sqrt[3]{3y}\right)\left(\sqrt[3]{9y^2}\right)$

B. Applying

4. Expand and simplify.

(a) $\left(4-\sqrt{5}\right)\left(2+\sqrt{5}\right)$

(b) $\left(\sqrt{2}+\sqrt{3}\right)\left(2\sqrt{2}-\sqrt{3}\right)$

(c) $\left(3\sqrt{5}-1\right)^2$

(d) $\left(\sqrt{6}+\sqrt{3}\right)^2$

5. Write the conjugate of each expression, then multiply the expression by its conjugate. State the restriction in (c).

(a) $6-\sqrt{10}$

(b) $2\sqrt{5}+\sqrt{3}$

(c) $\sqrt{x}-4$

6. (a) A rectangle measures $\left(5+\sqrt{3}\right)$ cm by $\left(5-\sqrt{3}\right)$ cm. Determine its area and its perimeter.

(b) A square has sides of length $\left(\sqrt{8}+\sqrt{2}\right)$ m. Determine its area. (Simplify the side first.)

7. The product $\left(\sqrt{5}-\sqrt{3}\right)\left(\sqrt{5}+\sqrt{3}\right)$ is equal to

A. 2 **B.** $\sqrt{2}$ **C.** 8 **D.** $2-2\sqrt{15}$

8. Numerical Response The product $\left(3+\sqrt{2}\right)\left(5-\sqrt{2}\right)$ can be written in the form $a+b\sqrt{2}$. The value of $a+b$ is _____.

C. Stretch

9. Show that $\left(\sqrt{x+1}+\sqrt{x}\right)\left(\sqrt{x+1}-\sqrt{x}\right)=1$ for every $x \geq 0$.

10. Determine k if $\left(\sqrt{k}+3\right)\left(\sqrt{k}-3\right)=7$.

11. Expand and simplify $\left(1+\sqrt{2}\right)^3$.

12. Show that $x=2+\sqrt{3}$ is a solution of $x^2-4x+1=0$.

Extra Practice 5: Dividing Radicals and Rationalizing Monomial Denominators

A. Skills

1. Divide and write in simplest form.

(a) $\dfrac{\sqrt{99}}{\sqrt{11}}$ (b) $\dfrac{18\sqrt{35}}{6\sqrt{5}}$ (c) $\dfrac{\sqrt[3]{324}}{\sqrt[3]{4}}$

(d) $\dfrac{9\sqrt{80}}{3\sqrt{10}}$ (e) $\dfrac{-20\sqrt{66}}{5\sqrt{11}}$ (f) $\dfrac{4\sqrt[3]{375}}{2\sqrt[3]{3}}$

2. Simplify.

(a) $\dfrac{\sqrt{50}}{\sqrt{18}}$ (b) $\dfrac{6\sqrt{20}}{4\sqrt{45}}$ (c) $\dfrac{\sqrt[3]{24}}{\sqrt[3]{81}}$

3. Rationalize the denominator and simplify.

(a) $\dfrac{3}{\sqrt{7}}$ (b) $\dfrac{10}{\sqrt{5}}$ (c) $\dfrac{\sqrt{5}}{\sqrt{6}}$

(d) $\dfrac{9}{2\sqrt{3}}$ (e) $\dfrac{20}{\sqrt{8}}$ (f) $\sqrt{\dfrac{7}{10}}$

(g) $\dfrac{1}{\sqrt[3]{7}}$ (h) $\dfrac{9}{\sqrt[3]{3}}$ (i) $\dfrac{2}{\sqrt[4]{2}}$

4. Divide each term of the numerator by the denominator.

(a) $\dfrac{\sqrt{30}-\sqrt{20}}{\sqrt{5}}$

(b) $\dfrac{12\sqrt{21}+18\sqrt{14}}{6\sqrt{7}}$

B. Applying

5. Simplify fully.

(a) $\dfrac{8-\sqrt{12}}{\sqrt{2}}$

(b) $\dfrac{\sqrt{18}+\sqrt{50}}{\sqrt{8}}$

(c) $\dfrac{9-3\sqrt{6}}{\sqrt{3}}$

6. Simplify. State the restrictions on the variable.

(a) $\dfrac{\sqrt{48a^5}}{\sqrt{3a}}$

(b) $\dfrac{10}{\sqrt{5x}}$

(c) $\sqrt{\dfrac{7}{3y}}$

(d) $\dfrac{9}{\sqrt[3]{3x^2}}$

7. A square has a diagonal of $4\sqrt{3}$ m. Its side is the diagonal divided by $\sqrt{2}$. Determine the exact side length and the area.

8. A rectangular patio has an area of $24\sqrt{6}$ m^2 and a width of $2\sqrt{3}$ m. Determine the exact length and perimeter.

9. Solve $x\sqrt{10}=\sqrt{20}+\sqrt{45}$. Write x in simplest form.

10. Which expression is in simplest form?

A. $\sqrt{\dfrac{2}{5}}$

B. $\dfrac{3}{\sqrt{2}}$

C. $\dfrac{\sqrt{6}}{3}$

D. $\dfrac{\sqrt{12}}{2}$

11. Numerical Response In simplest form, $\frac{\sqrt{3}}{\sqrt{8}} = \frac{\sqrt{a}}{b}$, where a and b are whole numbers. The value of $a + b$ is _____.

C. Stretch

12. Simplify $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{8}} + \frac{1}{\sqrt{18}}$.

13. Rationalize the denominator of $\frac{6}{\sqrt[3]{12}}$ using the smallest possible multiplier.

14. Show that $\sqrt{\frac{x}{y}} = \frac{\sqrt{xy}}{y}$ for $x \geq 0$, $y > 0$, and use it to write $\sqrt{\frac{11}{6}}$ in simplest form.

Extra Practice 6: Rationalizing Binomial Denominators with Conjugates

A. Skills

1. Write the conjugate of each binomial, and find the product of the binomial and its conjugate.

(a) $6 - \sqrt{7}$ (b) $\sqrt{13} + \sqrt{5}$ (c) $3\sqrt{3} - 2\sqrt{2}$

2. Rationalize the denominator and simplify.

(a) $\dfrac{5}{\sqrt{6}+1}$ (b) $\dfrac{2}{3-\sqrt{7}}$ (c) $\dfrac{8}{\sqrt{11}-\sqrt{3}}$

(d) $\dfrac{\sqrt{3}}{\sqrt{3}+3}$ (e) $\dfrac{6}{2\sqrt{2}+\sqrt{5}}$ (f) $\dfrac{\sqrt{6}}{\sqrt{3}-\sqrt{2}}$

3. Rationalize the denominator and simplify.

(a) $\dfrac{\sqrt{6}-2}{\sqrt{6}+2}$ (b) $\dfrac{1+\sqrt{3}}{2-\sqrt{3}}$ (c) $\dfrac{3\sqrt{2}-\sqrt{3}}{\sqrt{2}+\sqrt{3}}$

B. Applying

4. Rationalize the denominator. State the restrictions on the variable.

(a) $\dfrac{5}{\sqrt{x}+1}$ (b) $\dfrac{\sqrt{y}+3}{\sqrt{y}-3}$ (c) $\dfrac{2\sqrt{x}}{\sqrt{x}-\sqrt{3}}$

5. Simplify.

(a) $\dfrac{1}{\sqrt{2}+1}+\dfrac{1}{\sqrt{2}-1}$

(b) $\dfrac{6}{\sqrt{3}}-\dfrac{4}{\sqrt{3}-1}$

6. Simplify $\dfrac{\sqrt{18}-\sqrt{8}}{\sqrt{2}+1}$.

7. A rectangle has an area of 3 m^2 and a length of $(\sqrt{6}+\sqrt{3})$ m. Determine its exact width and perimeter.

8. Which expression is a rational number?

A. $\dfrac{1}{\sqrt{5}-2}$ **B.** $(\sqrt{5}+2)(\sqrt{5}-2)$ **C.** $(\sqrt{5}+2)^2$ **D.** $\dfrac{\sqrt{5}}{\sqrt{5}+2}$

9. Numerical Response The expression $\dfrac{1}{\sqrt{10}-3}$ can be written as $a+\sqrt{b}$, where a and b are whole numbers. The value of $a+b$ is _____.

C. Stretch

10. Rationalize the denominator of $\dfrac{1}{\sqrt{x+1}-\sqrt{x}}$, $x \geq 0$.

11. Rationalize the denominator of $\dfrac{1}{1+\sqrt{2}+\sqrt{3}}$. (Hint: group it as $(1+\sqrt{2})+\sqrt{3}$ and multiply by $(1+\sqrt{2})-\sqrt{3}$; then rationalize again.)

12. Use $(a-b)(a^2+ab+b^2)=a^3-b^3$ with $a=\sqrt[3]{2}$ and $b=1$ to rationalize the denominator of $\dfrac{1}{\sqrt[3]{2}-1}$.

Extra Practice 7: Unit 2 Review — Practice Test

Give every answer exact and in simplest form.

A. Multiple Choice

1. $\sqrt[3]{-128}$ in simplest form is

A. $-2\sqrt[3]{16}$ **B.** $-4\sqrt[3]{2}$ **C.** $-8\sqrt[3]{2}$ **D.** $4\sqrt[3]{2}$

2. The numbers $3\sqrt{11}$, $7\sqrt{2}$, $4\sqrt{6}$ and $2\sqrt{26}$ are ranked from least to greatest. The third number in the list is

A. $3\sqrt{11}$ **B.** $7\sqrt{2}$ **C.** $4\sqrt{6}$ **D.** $2\sqrt{26}$

3. The expression $\sqrt{4x-10}$ is defined for

A. $x \geq \frac{5}{2}$ **B.** $x \leq \frac{5}{2}$ **C.** $x \geq -\frac{5}{2}$ **D.** $x \geq 10$

4. For $x, y \geq 0$, $\sqrt{32x^5y^2}$ in simplest form is

A. $4x^2y\sqrt{2x}$ **B.** $16x^2y\sqrt{2x}$ **C.** $4x^2y\sqrt{2}$ **D.** $4x^2\sqrt{2xy}$

5. $4\sqrt{3} \cdot 2\sqrt{6}$ in simplest form is

A. $6\sqrt{18}$ **B.** $24\sqrt{2}$ **C.** $6\sqrt{9}$ **D.** $48\sqrt{2}$

6. $\dfrac{\sqrt{60}+\sqrt{24}}{\sqrt{6}}$ simplifies to

A. $\sqrt{10}+2$ **B.** $\sqrt{10}+4$ **C.** $\sqrt{14}$ **D.** $2\sqrt{10}+2$

7. When $2\sqrt{5}-\sqrt{3}$ is multiplied by its conjugate, the result is

A. 17 **B.** 7 **C.** 23 **D.** $17-4\sqrt{15}$

8. $\dfrac{4}{\sqrt[3]{2}}$, written with a rational denominator, is

A. $2\sqrt[3]{2}$ **B.** $2\sqrt[3]{4}$ **C.** $4\sqrt[3]{4}$ **D.** $2\sqrt{2}$

9. A rectangle has length $\sqrt{50}$ cm and width $\sqrt{8}$ cm. Its exact perimeter is

A. $14\sqrt{2}$ cm **B.** 20 cm **C.** $7\sqrt{2}$ cm **D.** $2\sqrt{58}$ cm

10. Which number is rational?

A. $\sqrt{2}+\sqrt{8}$ **B.** $\sqrt{3}\cdot\sqrt{27}$ **C.** $\dfrac{\sqrt{5}}{\sqrt{10}}$ **D.** $\left(1+\sqrt{2}\right)^2$

B. Numerical Response

11. Numerical Response The expression $\sqrt{12}\left(\sqrt{6}+\sqrt{3}\right)-\sqrt{8}\left(3-\sqrt{2}\right)$ simplifies to a whole number. The whole number is _____.

12. Numerical Response $\left(2\sqrt{3}-1\right)^2$ can be written in simplest form as $a-b\sqrt{c}$. The value of $a+b+c$ is _____.

13. Numerical Response $\dfrac{5+\sqrt{3}}{2-\sqrt{3}}$ can be written in the form $a+b\sqrt{3}$. The value of ab is _____.

14. Numerical Response A square has an area of 90 cm^2. Its diagonal can be written in simplest form as $a\sqrt{b}$ cm. The value of ab is _____.

C. Written Response

15. Simplify. Assume $a \geq 0$ and $x > 0$.

(a) $\sqrt{18a^3}+a\sqrt{50a}-\sqrt{2a^3}$ (b) $\sqrt[3]{24x^4}\cdot\sqrt[3]{9x^2}$ (c) $\dfrac{6}{\sqrt{3x}}$

16. Rationalize the denominator and simplify.

(a) $\dfrac{\sqrt{10}}{\sqrt{5}+\sqrt{2}}$ (b) $\dfrac{3\sqrt{2}-4}{3\sqrt{2}+4}$

17. A rectangle has an area of 8 cm^2 and a width of $\left(\sqrt{5}-1\right)$ cm.

(a) Show that the length of the rectangle is $\left(2\sqrt{5}+2\right)$ cm.

(b) Determine the exact perimeter of the rectangle.

(c) A square has the same area as the rectangle. How much greater is the perimeter of the rectangle than the perimeter of the square? Give an exact answer.

18. (a) Show that $\dfrac{1}{3+2\sqrt{2}} + \dfrac{1}{3-2\sqrt{2}}$ is a rational number.

(b) A student says, "The product of two irrational numbers is always irrational." Use a radical example to show the student is wrong.

(c) Simplify $\sqrt{\dfrac{7}{18}}$ and explain which simplest-form rule it breaks before you simplify it.